

Making the Genomic Pathologist's Day

10x Better

● Abhimanyu Verma

Chief Technology Officer

● Shubham Nayyar

Head of Tech - DSE & Core Platform

● Antonio Ridi

Sr. Manager - Tech Platform Lead

● Jesus Garcia

Distinguished Software Engineer

A category-defining software company

On a mission to democratize data-driven medicine, together

2 M+

**Genomic profiles
analyzed**

~1,000 analyses per day

800+

**Healthcare
institutions**

70+ countries worldwide

2,000+

**Scientific
publications**

as of March 2025

2011

Founded

IPO NASDAQ:SOPH · Jul 2021



Democratizing Data-Driven Medicine, Together.

OUR MISSION

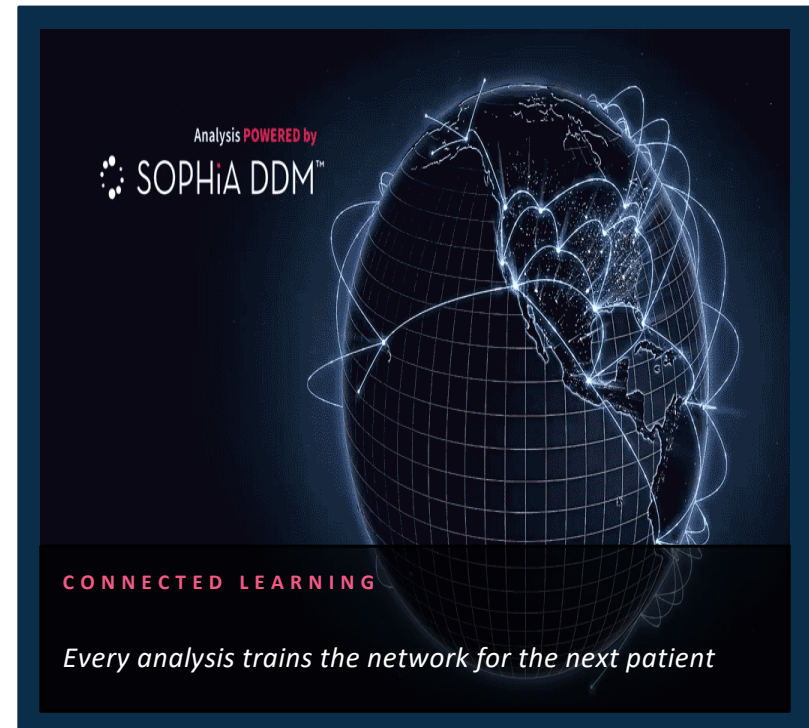
A patient in Istanbul today,

helps a patient in Boston tomorrow

Expand access to **data-driven medicine** by using **AI** to deliver world-class care to patients **across the globe.**

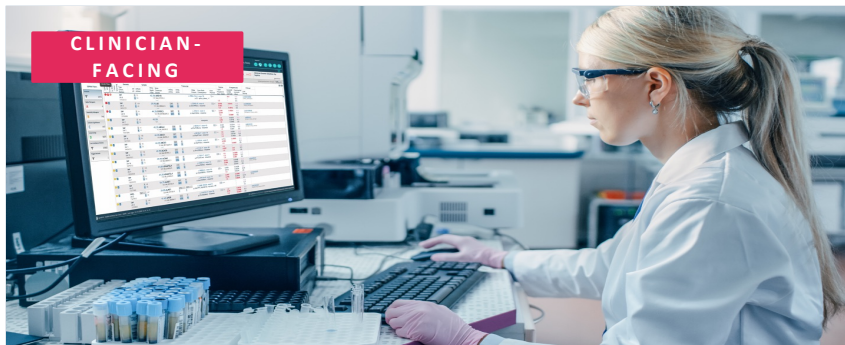
AI Agents make this vision actionable -

one variant annotation at a time.



Two professionals, one workflow

From the lab bench to the patient bedside



Lab Professionals

Genomic pathologists · Clinical geneticists ·
Bioinformaticians

ANALYSIS
REPORT



Healthcare Professionals

Oncologists · Rare disease clinicians · Treatment teams

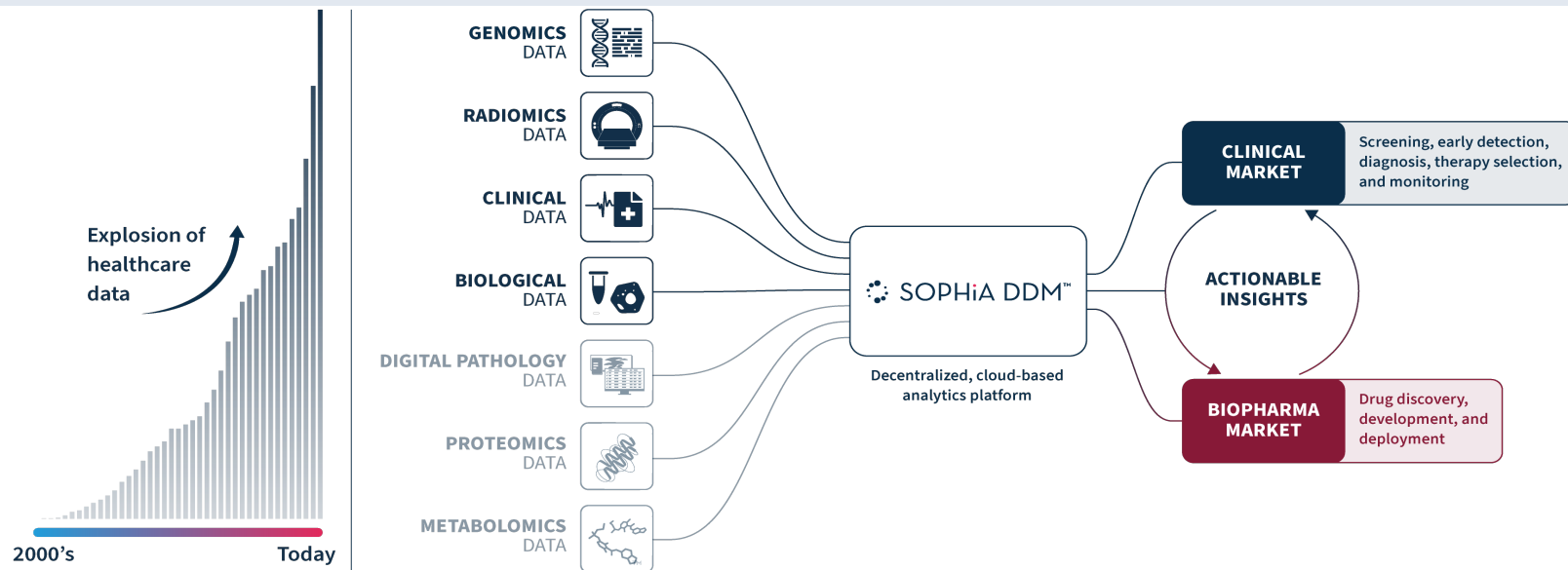
What is the root cause of the disease?

Oncology

Rare disorders

SOPHiA DDM™ - a decentralized analytics platform

Computes complex multimodal data into actionable insights for Clinical & BioPharma customers



AI-POWERED

AI/ML

multimodal analytics platform

UNIVERSAL

Any

sequencer or enrichment tech

SCALABLE

1,000+

analyses run per day

02

Genomics Crash Course

DNA → Variants → Disease

What you need to know in 3 slides

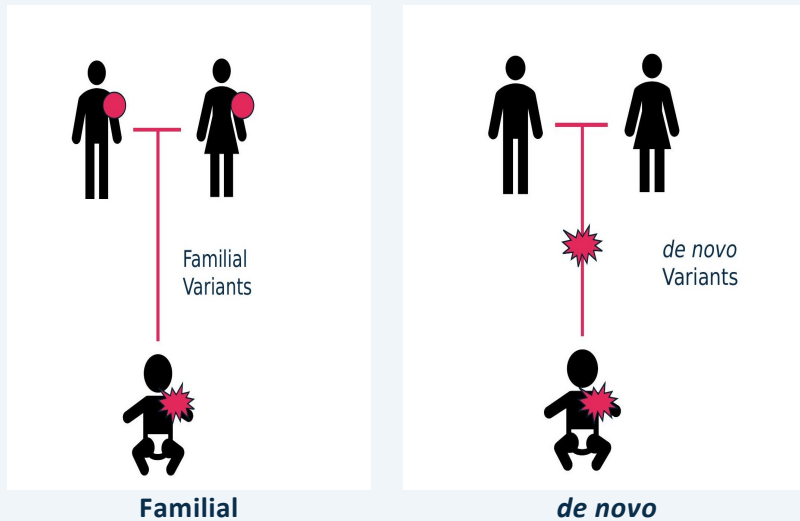
Variants happen two ways

A "variant" is a difference between a patient's DNA and the reference genome

INHERITED

Germline / Constitutional

Present in every cell of the body



ACQUIRED

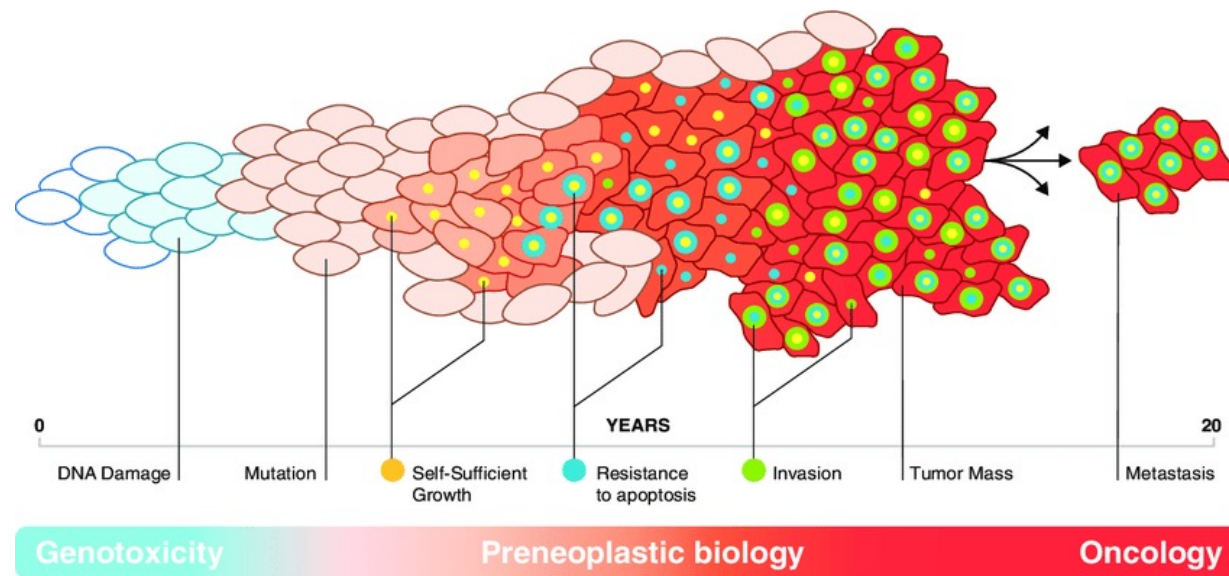
Somatic

Develop during a person's lifetime

- Hereditary risk factors
- UV exposure
- Smoking / tobacco use
- Carcinogen exposure
- Pathogenic infections

When variants stack up, a tumor emerges

Genotoxicity → Preneoplastic biology → Oncology - a process that can take 20 years



Adapted from Salk et al., 2019

1

DNA damage

Variants accumulate

2

Self-sufficient growth

Cells stop responding to control signals

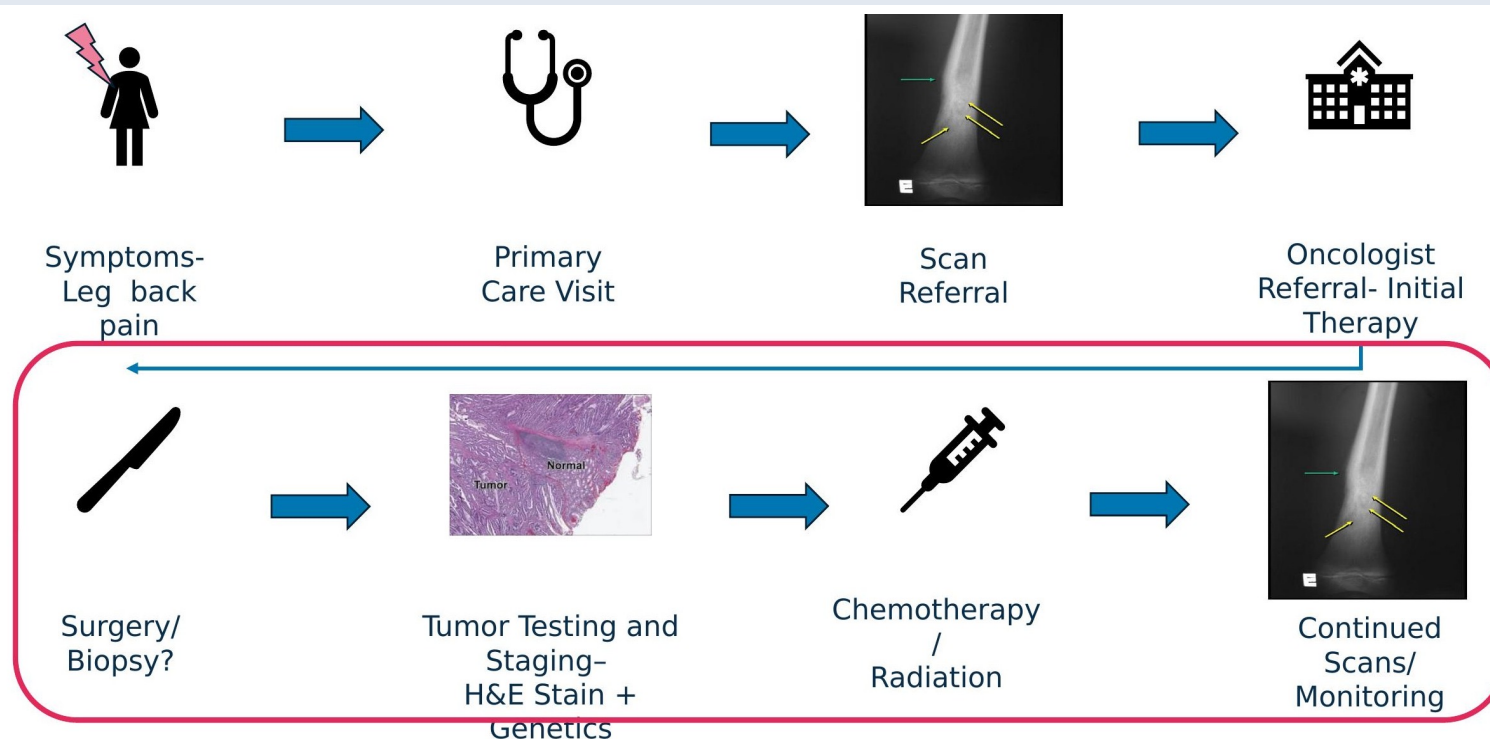
3

Tumor mass

Invasion and, eventually, metastasis

A cancer diagnosis can take months

Genomic testing sits at the centre - guiding biopsy interpretation, staging, and therapy choice



Where SOPHiA DDM™ steps in: tumor testing & staging - H&E stain + Genetics.

03

A Day in the Life of a Pathologist

What is she expected to deliver?

The expert at the centre of every diagnosis

Meet a genomic pathologist

The expert who turns sequencing data into a clinical diagnosis



Dr. M.

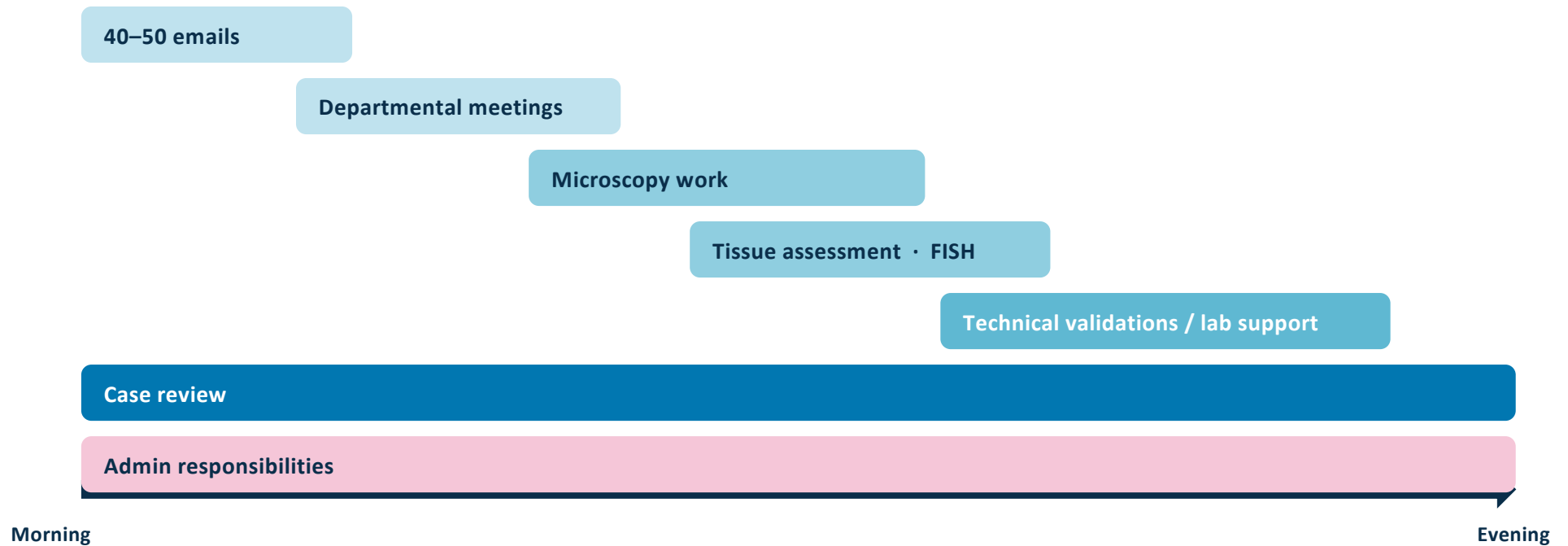
*Hematopathology &
Molecular Genetic Pathology*

PRIMARY RESPONSIBILITIES

- **Molecular Service** Reviews molecular data · Writes clinical reports
- **Hematopathology Service** Reviews H&E slides · Writes clinical reports
- **Departmental Reviews** Sign-off and quality oversight
- **Lab Director Duties** Wetlab support · Technical validations
- **Administrative** Budget · Hiring · Process

Many roles, one timeline

On the molecular / hematopathology service, the day stacks up fast



Case review runs the full day - each case is a real patient awaiting a decision.

Complexity is exploding around the pathologist

Therapies, knowledge and data are all growing faster than expert capacity

GROWING THERAPIES

35%

of new FDA-approved therapies in 2021 reference a diagnostic test

47%

of approved personalized medicines were oncology

PMC, Personalized Medicine at FDA, 2021

GROWING KNOWLEDGE

1 year

doubling time of medical knowledge today - down from 50 years in 1950

95

new studies published every hour

Densen, Trans Am Clin Climatol Assoc 2011

GROWING DATA

7.5 ZB

of healthcare data needs a long-term home in 2025 (581% increase from 2019)

100 GB+

of raw data per genomic sample

Seagate Data Age 2025; Harmony Healthcare IT

Four pressures pulling on every case

Each one is a place where AI can lift cognitive load - without replacing judgment

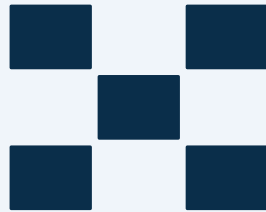
1



Myriad Responsibilities

Reviewing slides, signing reports, lab management, admin - all in one day.

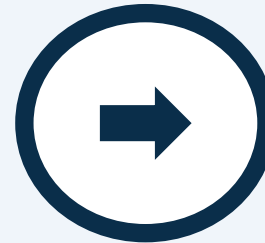
2



Disparate Technologies

Microscope, FISH, NGS, IHC - each in its own system, each with its own UI.

3



Evolving Literature

New variants and guidelines published faster than any individual can read.

4



Improve Outcomes

The bar keeps rising. Patients deserve the best diagnosis, every time.

NOW IMAGINE THIS

8:00 AM
Monday

She arrives at the lab.

Genomic analyses ran overnight.

Pending her interpretation and *patients waiting for a diagnosis.*

Where does she start?

What she actually does, case by case

Inside SOPHiA DDM™ Gen2 - from a sequenced sample to a signed report

1

Open the Genomics Homepage

See backlog of pending interpretations

2

Find the case in the Finder

Filter by run, sample, or status

3

Open the Interpretation

Quality, gene count, oncogenicity overview

4

Gene View

Which genes have called variants?

5

Variant Table

Filter / classify · SNV·INDEL · Fusions
· CNV

6

Flag & write conclusion

Build the report, add interpretation text

7

Sign off · or send for review

Close the case and move to the next one

SEE IT IN ACTION

[LIVE DEMO]

→ Appendix slides A1–A8 give a full screen-by-screen walkthrough

Help pathologists make better decisions, faster

Caseloads are growing. Expert pathologist time is finite. AI can surface the right information at the right moment.

Four directions to inspire - pick one, combine them, or surprise us:

1

SUMMARY

Smart Case Summary

AI-generated 4–5 bullet overview on the analysis index page - fast triage before deep review.

2

PRE-FLAG

AI Variant Pre-flagging

Pull data via VQS API · LLM ranks which variants deserve priority and explains why.

3

BROWSER

Natural Language Browser Agent

Chrome extension that navigates Gen2 from plain English ("show me tier-1 variants for this case").

4

BATCH

Task Management for Pathologist

Organize · let the pathologist prioritize and work on tasks b/w multiple pending interpretations.

You don't have to pick from the list - surprise us with something we didn't think of.

What you have access to

Real environment, real data shapes, real mentors - for the full weekend

ENVIRONMENT

SOPHiA DDM™ Gen2

- Team accounts on the demo environment
- Pre-loaded analyses
- Pipelines already run - interpretations & reports
- Backup environment available if needed
- **URL & Credentials sent to each team in the student pack**

API · CLI

VQS API

- The data layer powering Gene & Variant views
- Variant data · oncogenicity · interpretation results
- Quickstart doc with auth + curl examples
- **Refer to API details & Guidance in the student pack**

SUPPORT

On-Site Mentors

- Shubham Nayyar · Saturday + Sunday
- Antonio Ridi · Saturday + Sunday
- Jesus Garcia · Saturday + Sunday
- WhatsApp / Discord- come find us
- **Also refer to the recorded platform Demo in the student pack**

How we'll evaluate your build

Equal weight · clinical reality matters more than sparkle

01

User Impact

Would a real genomic pathologist find this useful in daily practice? Does it address a real workflow moment?

02

Technical Execution

Does it actually work? Does it integrate with the live Gen2 environment or VQS API - not just a mock?

03

UX Quality

Would a time-pressed expert adopt this? Is the interaction intuitive and does it fit the workflow naturally?

04

Innovation

Did you find an angle SOPHiA hadn't thought of? Is there genuine creativity in the approach?

Open the laptop. Find your first case.

Everything you need is in the student pack - credentials, API quickstart, glossary, ground rules

IN THE PACK

- **Team account credentials** Username · password · environment URL
- **3-Layer Data Model** Run → Analysis → Interpretation · how to navigate
- **VQS API Quickstart** Base URL · auth · curl examples to get a variant list
- **Data Glossary** Variant · SNV/INDEL · CNV · Fusion · VUS · ABCD
- **Ground Rules** No real patient data · no triggering live pipelines
- **Support contact** WhatsApp / Slack for the SOPHiA team

WE'RE HERE FOR YOU

The SOPHiA team - on-site all weekend

We're genuinely curious to see what you build. Come find us.



Shubham Nayyar

Head of Tech - DSE & Core Platform



Antonio Ridi

Senior Manager- Technology Platform Lead



Jesus Garcia

Distinguished SDE

FIND US IN PERSON

We'll be there all weekend at ETH PbLabs. Come ask us anything.

WhatsApp / Slack details in your student pack.